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Linear Algebra

7Week [16 October - 22 October]

Midterm

Started on	2023-10-13 19:01
State	Finished
Completed on	2023-10-13 20:44
Time taken	1 hour 43 mins
Grade	Not yet graded

Question 1

Partially correct

Mark 0.75 out of 1.00

Match the sentences' halves

The system

$$\begin{cases} x + y + z = 7 \\ x + y - z = 6 \\ x + y = 4 \end{cases}$$

is an inconsistent sistem



A 5×6

matrix

has six columns



$$a_1x_1 + a_2x_2 + a_3 = 0$$

is a linear equation with two variables



Solution of the equation is

a set of numbers can be substituted into



Your answer is partially correct.

You have correctly selected 3.

The correct answer is: The system

$$\begin{cases} x + y + z = 7 \\ x + y - z = 6 \\ x + y = 4 \end{cases} \rightarrow \text{is an}$$

inconsistent sistem, A 5×6 matrix $\rightarrow$ has six columns,

$a_1x_1 + a_2x_2 + a_3 = 0$ $\rightarrow$ is a linear system with two variables,

Solution of the equation is $\rightarrow$ a set of numbers can be substituted into equation and makes it identity

Question 2
Not answered
Not graded

Consider the following system of equations:

$$\begin{cases} 5x + y + z = 2 \\ x + 5y + 5z = 0 \\ x + 5y + 10z = 3 \end{cases}.$$

(a) (1 point) Write the matrix A and the vector $\vec{b}$ such that \$

$$A \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \vec{b}.$$

(b) (2 points) Use Gaussian elimination, put the augmented matrix $(A|\vec{b})$ and find the row echelon form.

(c) (2 points) Continuing in (b) put the augmented matrix $(A|\vec{b})$ in reduced row echelon form that is the result will be

$$\left(\begin{array}{ccc|c} 1 & 0 & 0 & a \\ 0 & 1 & 0 & b \\ 0 & 0 & 1 & c \end{array} \right), \text{ for some numbers } a, b, c.$$

(d) (1 points) Using result of (c) write the solution of the given system.

(e) (3 points) Find A^{-1} .

(f) (1 point) Solve the given system using the result of (e).

Attach your solution below.

Ignore the answer box.

Answer:

The correct answer is: 5.00

Question 3
Complete
Marked out of 10.00

Attach solution to the previous question here:

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Question **4**

Complete
Not graded

Solve the following system

$$\begin{cases} x - y + 4z = 6 \\ -4x + y = -5 \\ 4x + y + 4z = 1 \end{cases}$$

using Cramer's

rule.

Write the determinant of the matrix in the box below.

Attach your solution in the next question.

Answer:

The correct answer is: -44.00

Question **5**

Complete
Marked out of
9.00

Attach here solution of the previous question.

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